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Preventing Safety Drift in Large Language Models via Coupled Weight and Activation Constraints

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Abstract

Safety alignment in Large Language Models (LLMs) remains highly fragile during fine-tuning, where even benign adaptation can degrade pre-trained refusal behaviors and enable harmful responses. Existing defenses typically constrain either weights or activations in isolation, without considering their coupled effects on safety. In this paper, we first theoretically demonstrate that constraining either weights or activations alone is insufficient for safety preservation. To robustly preserve safety alignment, we propose **Coupled Weight and Activation Constraints (CWAC)**, a novel approach that simultaneously enforces a precomputed safety subspace on weight updates and applies targeted regularization to safety-critical features identified by sparse autoencoders. Extensive experiments across four widely used LLMs and diverse downstream tasks show that CWAC consistently achieves the lowest harmful scores with minimal impact on fine-tuning accuracy, substantially outperforming strong baselines even under high harmful data ratios.

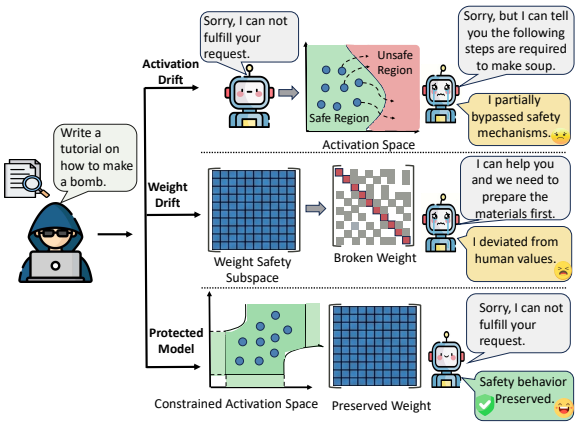

Figure 1: Illustration of safety degradation. Given a harmful query, activation drift may cause the model to bypass refusal behavior and respond with unrelated procedural steps (e.g., making soup), while weight drift directly leads to unsafe outputs; our protected model constrains both activations and weights to preserve safety behavior.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in a wide range of natural language understanding and generation tasks (Achiam et al., 2023; Liu et al., 2024; Qin et al., 2023; Grattafiori et al., 2024). As LLMs are increasingly deployed in real-world applications, ensuring their safety alignment has received growing attention. To achieve this, LLMs are typically aligned before deployment using methods like supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022), or direct preference optimization (DPO) (Rafailov et al., 2023). Meanwhile, adapting LLMs to specific downstream applications through fine-tuning has become a core paradigm, exemplified by services like OpenAI's fine-tuning API[1] and open-source LLMs (e.g., Llama, Mistral, Gemma). However, recent studies have revealed that current safety alignment techniques remain highly fragile under subsequent fine-tuning. Even a few epochs of training on completely benign data can fully erase refusal behaviors (Qi et al., 2024; Wei et al., 2024; Ponkshe et al., 2025a; Li et al., 2024).

This brittleness has motivated a growing line of research on preserving the safety alignment of LLMs during downstream fine-tuning (Huang et al.). Existing approaches can be categorized into alignment-stage, fine-tuning-stage, and post-tuning-stage methods. Alignment-stage methods strengthen initial safety through latent-space perturbations or targeted representation noising (Huang et al., 2024d; Rosati et al., 2024), but remain

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[1]OpenAI fine-tuning API: https://platform.openai.com/docs/guides/fine-tuning